

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460047.7209 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.84 +/- 2.22 [%]
Orbital Inclination (inc)	83.15 +/- 2.86
Airmass coefficient 1 (a1)	0.69 +/- 0.51
Airmass coefficient 2 (a2)	0.43 +/- 1.21
Scatter in the residuals of the lightcurve fit is	75.56 %
Best Comparison Star	None
Optimal Aperture	1.93
Optimal Annulus	3.75
Transit Duration (day)	0.053 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.196 ± 0.057 vs 0.1594 (deviation 0.55σ)
Duration fit vs published	0.053 d vs 0.0512 d (deviation 0.14σ)
Mid-time O-C	5.4 min
Depth SNR	2.9
Residual scatter	75.56 %

Light curve

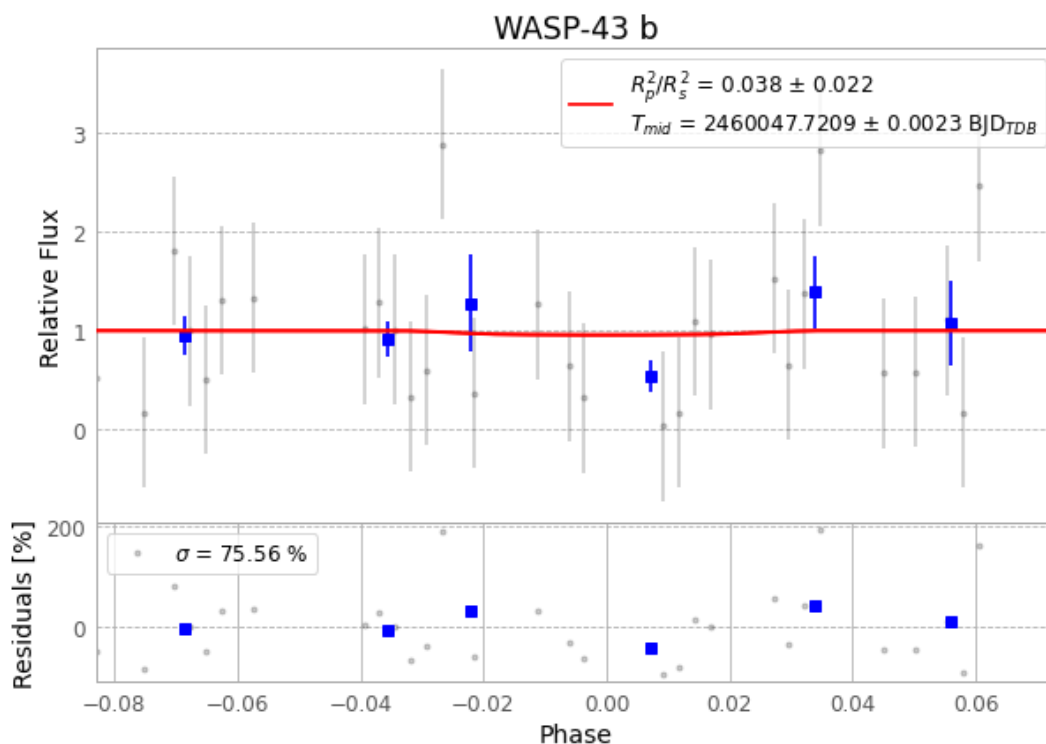


Figure 1 — FinalLightCurve_WASP-43 b_2023-04-12.png (SHA-256 2be1cfe4c3837010...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d75c11c9080a61ed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), WASP-43230413033315.FITS → WASP-43230413064215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230413.log (107675254f5a...), FinalLightCurve_WASP-43 b_2023-04-12.png (2be1cfe4c383...), FinalLightCurve_WASP-43 b_2023-04-12.csv (17089c17423a...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 08:42Z.